```
DENYWRITE, 3, 0) = 0x7f3b6c7d9000
mprotect(0x7f3b6c960000, 2097152, PROT_NONE) = 0
```

Here we see that multiple shared objects or libraries are being loaded.

```
socket(PF_INET, SOCK_DGRAM|SOCK_NONBLOCK, IPPROTO_IP) = 3
connect(3, {sa_family=AF_INET, sin_port=htons(53), sin_
addr=inet_addr("172.19.17.32")}, 16) = 0
poll([{fd=3, events=POLLOUT}], 1, 0)    = 1 ([{fd=3,
revents=POLLOUT}])
sendto(3, "\302\276\1\0\0\1\0\0\0\0\0\6google\3c
om\0\0\1\0\1", 28, MSG_NOSIGNAL, NULL, 0) = 28
poll([{fd=3, events=POLLIN}], 1, 5000)  = 1 ([{fd=3,
revents=POLLIN}])
ioctl(3, FIONREAD, [204])               = 0
recvfrom(3, "\302\276\201\200\0\1\0\1\0\0\0\0\6google\3com\
0\0\1\0\1\300\f\0\1"..., 1024, 0, {sa_family=AF_INET, sin_
port=htons(53), sin_addr=inet_addr("172.19.17.32")}, [16]) =
204
close(3)                                = 0
socket(PF_INET, SOCK_DGRAM, IPPROTO_IP) = 3
connect(3, {sa_family=AF_INET, sin_port=htons(1025), sin_
addr=inet_addr("74.125.239.41")}, 16) = 0
```

Further ahead are the *socket/connect* calls to the DNS server for the host lookup for google.com, which returns the IP address (74.125.239.41).

In this manner, by going through these lines, a great deal of the program and its interaction with the operating system can be learnt. Strace can be used to profile a program using the *-c* switch, as shown below:

```
$ strace -c ls /
% time     seconds usecs/call     calls    errors syscall
------ ----------- ----------- --------- --------- ----------
-----
 66.18    0.000045          23         2           getdents
 33.82    0.000023           2        12           open
  0.00    0.000000           0        10           read
  0.00    0.000000           0         1           write
0.00    0.000000           0        14           close
...
```

We can see here that the majority of the calls are made in *getdents* and *open* calls. *Getdents* is used to get directory entries. Check *man getdents* for more details.

Strace can also be used to trace a running process, as follows:

```
$ strace -p 31906
Process 31906 attached - interrupt to quit
select(0, NULL, NULL, NULL, {3, 702892}) = 0 (Timeout)
open(".lockfile", O_WRONLY|O_CREAT|O_TRUNC, 0777) = 10
flock(10, LOCK_EX|LOCK_NB)              = -1 EAGAIN (Resource
temporarily unavailable)
write(1, "Could not lock file, sleeping..."..., 33) = 33
select(0, NULL, NULL, NULL, {5, 0})     = 0 (Timeout)
```

Use the *-p* switch and pass in the process PID. We see that it is waiting for a file lock with file name *.lockfile*. Use *lsof* to check which program is holding up this file:

```
$ lsof | grep .lockfile
python    31935   rudiyar    4wW     REG                8,1
0   20452726 /home/rudayar/.lockfile
python    31942   rudiyar    3w      REG                8,1
0   20452726 /home/rudayar/.lockfile
python    31942   rudiyar    4w      REG                8,1
0   20452726 /home/rudayar/.lockfile
python    31942   rudiyar    5w      REG                8,1
0   20452726 /home/rudayar/.lockfile
python    31942   rudiyar    6w      REG                8,1
0   20452726 /home/rudayar/.lockfile
```

Run Strace on this again, to see what it is doing:

```
$ strace -p 31935
Process 31935 attached - interrupt to quit
select(0, NULL, NULL, NULL, {23, 302837}^C <unfinished ...>
Process 31935 detached
```

The process is just sleeping (the *select* call) after acquiring the lock. This was a test program, but a real application might be doing, or be stuck on, some other task.

I usually use the following switches to run Strace:

```
$ strace -Tfvto strace.out <application>

f  -  follow forked processes
v  -  to be more verbose
T  -  time spent in each system call (helps you to identify
latency issues)
t  -  adds timestamp to each call
o  -  output file to write to
```

Strace is a useful tool for systems administrators to understand program behaviour, without resorting to a debugger. It is a must for performance or latency analysis. It also serves as a good learning and teaching tool, as it provides valuable insights on how the underlying system works.

Next time you want to know why a program hangs, or is taking too long to execute, use Strace. END

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